Predictive Churn Analysis

Machine Learning Classification for Customer Retention Risk

Objective: Build a Logistic Regression pipeline to predict customer churn probability and export customer-level risk scores.

1. Dataset & Preprocessing

The sample contains 1,200 customer records. Numerical features are MinMax scaled and categorical features are One-Hot encoded. Missing values are handled inside the preprocessing pipeline.

2. Train-Test Split

The dataset is divided into 80% training data and 20% testing data using stratification and a fixed random seed.

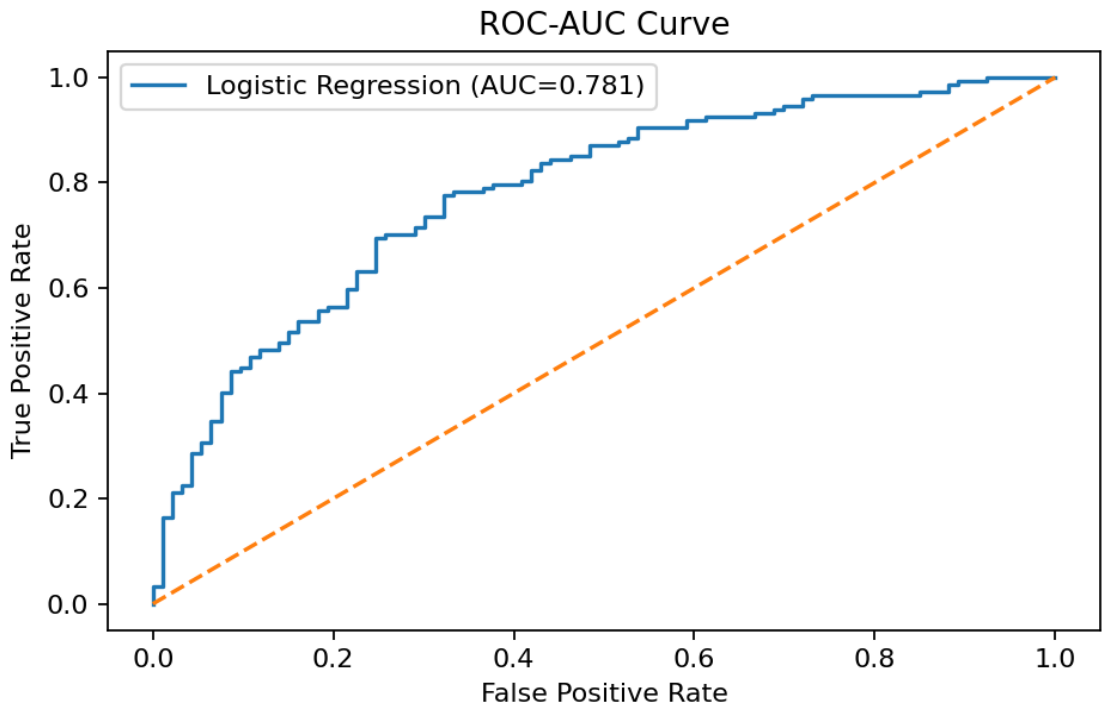
3. Model

Logistic Regression with class balancing is used for binary churn classification. The model also produces a churn probability for every customer.

4. Evaluation Results

Metric	Test Result
Precision	0.791
Recall	0.721
F1-Score	0.754
ROC-AUC	0.781

5. ROC-AUC Curve



6. Risk Score Output

Each customer receives a churn probability. Risk bands are Low (<0.40), Medium (0.40–0.69), and High (≥0.70). The full predictions are exported to `churn_risk_predictions.csv`.

7. Business Use

High-risk customers can be prioritized for retention campaigns, customer support outreach, or personalized offers.

8. Conclusion

The project demonstrates a complete ML classification pipeline from preprocessing and model training to evaluation and customer-level churn risk prediction.